

BMET3997/9997
Physiological signal sensing and processing
Major project: HRV parameters



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Philip de Chazal

Semester 1, 2026



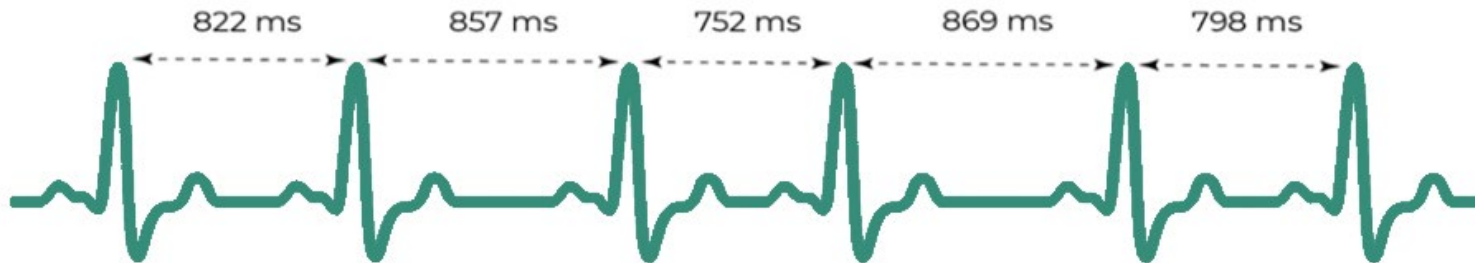
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custodianship and law of the Country on which
the University of Sydney campuses stand.
We pay our respects to those who have cared
and continue to care for Country.



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Measurement of HRV

- Heart rate variability is simply a measure of the variation in time between each heartbeat.
- It can be measured using the ECG



Steps for the HRV calculation

- 1. QRS detection**
 - 2. Identifying the non-normal beats**
 - 3. Calculate the RR intervals between the successive normal beats**
 - 4. Calculate the HRV parameters (see next slides)**
- Issue: some HRV methods assume that the RR intervals are contiguous (ie. no gaps). Ectopic beats cause issues!**

HRV time domain features

Parameter	Calculation
Mean RR-interval	Average value of the RR intervals
SDNN: Standard deviation of the RR-interval	Standard deviation of the RR intervals
Serial correlation of the RR intervals	Correlation of the RR intervals with the RR intervals shifted by fixed number of beats
NN50	The number of pairs of adjacent RR-intervals where the first RR-interval differs from the second RR-interval by more than 50 ms;
pNN50	NN50 measure divided by the total number of RR-intervals *100
SDSD	The standard deviation of the differences between adjacent RR-intervals
RMSSD	The square root of the mean of the squares of differences between adjacent RR-intervals
HRV triangular index	Total number of all RR intervals divided by the maximum of the density distribution of the RRs

HRV time domain features – Matlab code

Assume we have an RR vector with values scaled in seconds.e.g `RR=[0.85, 0.90, 1.00, 0.98, 1.1...]`'

Parameter	Matlab
Mean RR-interval	<code>meanRR=mean (RR) ;</code>
SDNN: Standard deviation of the RR-interval	<code>stdRR=std (RR)</code>
Serial correlation of the RR intervals at 1 interval shift	<code>L=length (RR) ; temp=xcorr (RR,'coeff') ; Corr=temp (L+1) ;</code>
NN50	<code>NN50=sum (diff (RR)>0.05 diff (RR)<-0.05) ;</code>
pNN50	<code>pNN50=NN50/ (length (RR)) ;</code>
SDSD	<code>SDSD=std (diff (RR)) ;</code>
RMSSD	<code>RMSSD=sqrt (mean (diff (RR).^2)) ;</code>
Triangular index	<code>[~,count]=mode (RR) ; TriI=length (RR) /count ;</code>

HRV time domain features – Python code

Assume we have an RR vector with values scaled in seconds.e.g `RR=[ 0.85, 0.90, 1.00, 0.98, 1.1]`

```
import numpy as np
from scipy import stats
```

Parameter	Python
Mean RR-interval	<code>meanRR=np.mean(RR)</code>
SDNN: Standard deviation of the RR-interval	<code>stdRR=np.std(RR)</code>
Serial correlation of the RR intervals at 1 interval shift	<code>L=length(RR); temp=xcorr(RR,'coeff'); Corr=temp(L+1);</code>
NN50	<code>NN50=np.sum((np.diff(RR) > 0.05) (np.diff(RR) < -0.05))</code>
pNN50	<code>pNN50=NN50/len(RR)</code>
SDSD	<code>SDSD=np.std(np.diff(RR))</code>
RMSSD	<code>RMSSD=np.sqrt(np.mean(np.diff(RR)**2))</code>
Triangular index	<code>mode_value, count = stats.mode(RR) TriI = len(RR) / count[0]</code>

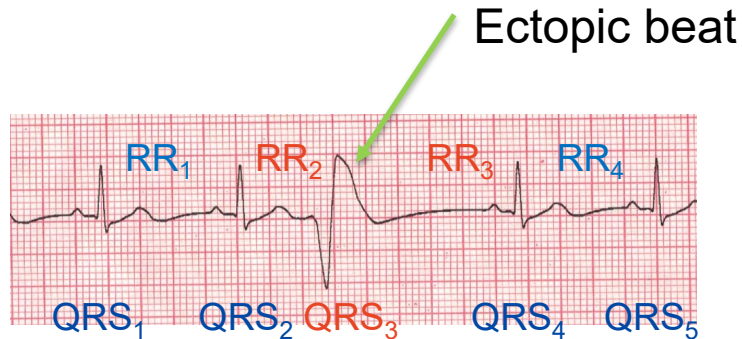
Ectopic beats and noise

- **Important: HRV is normal-normal beat variability**
 - Ideally, remove ectopic beats first as can heavily influence HRV
 - The choice of ectopic detector can influence HRV
- Each ectopic beats affects 2 RR intervals



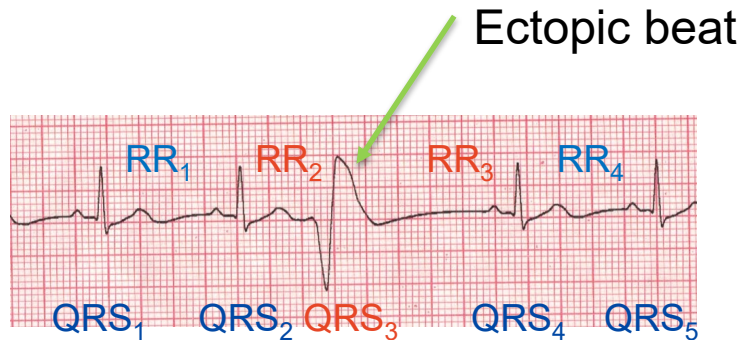
RR interval (ms)							St. dev. (ms)
uncorrected	856	652	1235	870	905	632	218
corrected	856	-	-	870	905	-	25

Calculation of HRV parameters in the presence of non-normal beats- Matlab



- **Each non-normal beat affects two RR intervals**
 - In this case the ectopic beat **QRS₃** affects **RR₂** and **RR₃**
- **A convenient way in Matlab is to assign the QRS times associated with non-normal beats to NaN.**

Calculation of HRV parameters in the presence of non-normal beats- Matlab



For above say

```
>> QRS=[1, 1.95, 2.23, 3.94, 4.87] %Time of each QRS in seconds
```

Process including the ectopic beats

```
%Calculate RR intervals
```

```
>> RR=diff(QRS);
```

```
%Calculate mean and standard deviation of RR intervals
```

```
>> meanRR=mean(RR); stdRR=std(RR)
```

```
>> [meanRR,stdRR]
```

```
ans =
```

```
0.9675    0.5847
```

Process ignoring the ectopic beat for QRS₃

```
>> QRS[3]=nan; %Set the ectopic beat to nan
```

```
%Calculate RR intervals
```

```
>> RR=diff(QRS);
```

```
%Calculate mean and standard deviation of RR intervals
```

```
>> meanRR=mean(RR,'omitnan'); stdRR=std(RR,'omitnan') %Use the option 'omitnan' in the calculating function
```

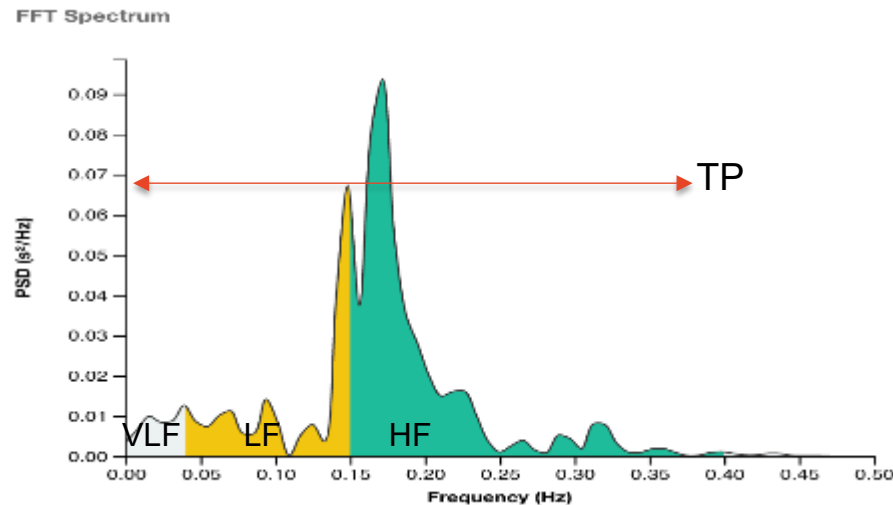
```
>> [meanRR,stdRR]
```

```
ans =
```

```
0.9400    0.0141
```

Much smaller standard deviation as we have ignored the very long and very short RR intervals caused by the ectopic beat

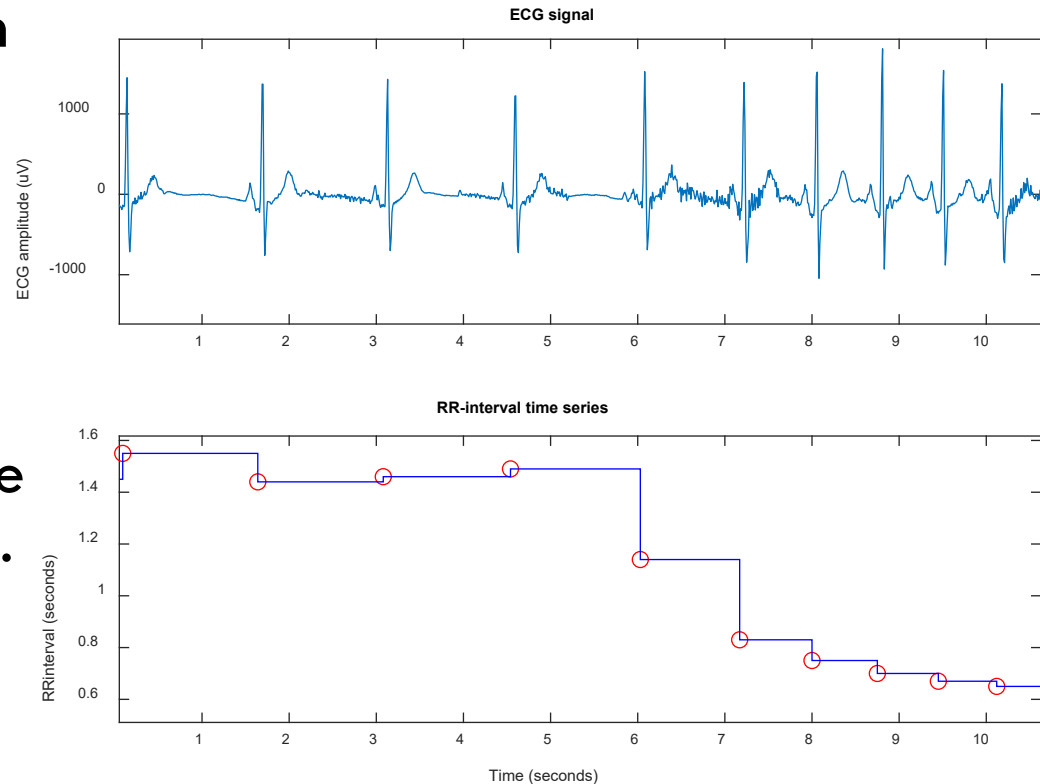
HRV frequency domain features



Parameter	Calculation
VLF	Power in the very-low-frequency range i.e ≤ 0.04 Hz for short term recordings, and 0.003-0.04Hz for 24-hour recordings
LF	Power in the low-frequency range. 0.04 - 0.15 Hz
HF	Power in the high-frequency range 0.15-0.4 Hz
TP	Total power ≤ 0.4 Hz. It is equal to the variance of NN intervals over the selected segment
LF/HF	The ratio of between LF and HF power

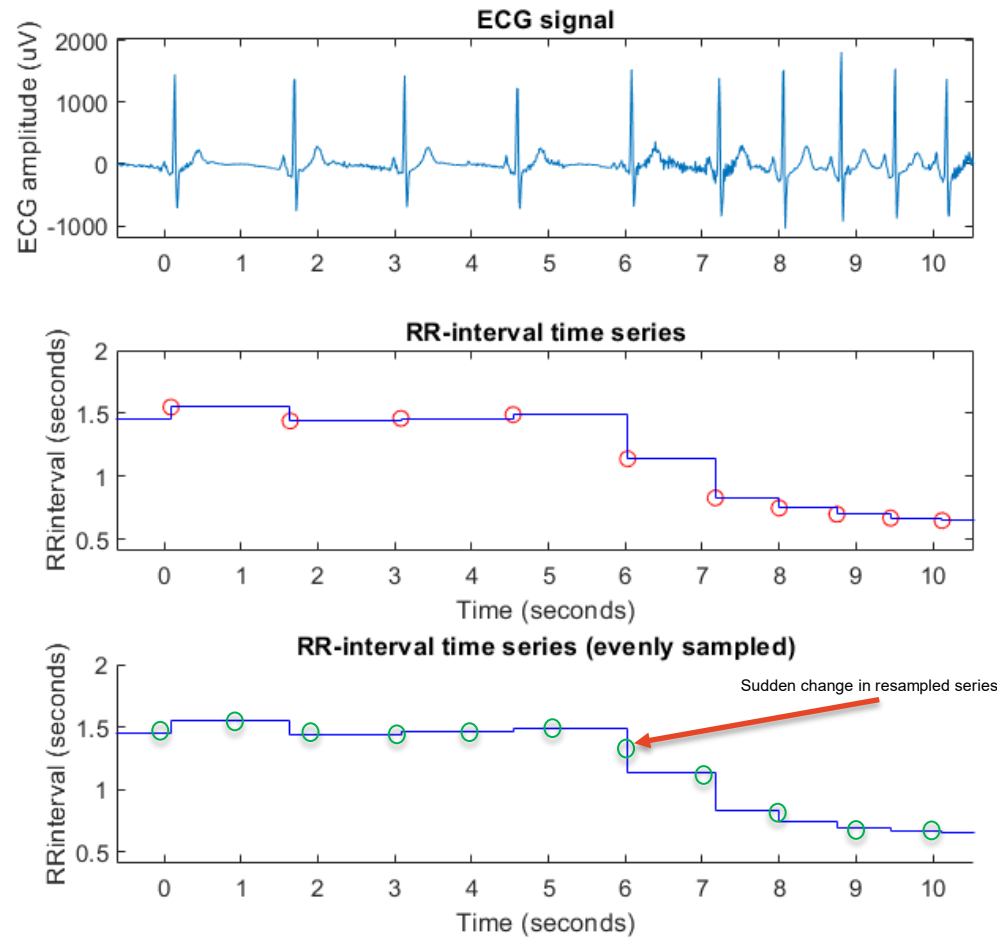
Calculating PSD of RR intervals

- Previous slides have shown that the PSD of the HRV can reveal some very useful information.
- How do we calculate this spectrum?
 - QRS points are unevenly sampled, hence the RR time series is unevenly sampled.
 - We can't apply DFT to RR time series as it assumes evenly sampled series



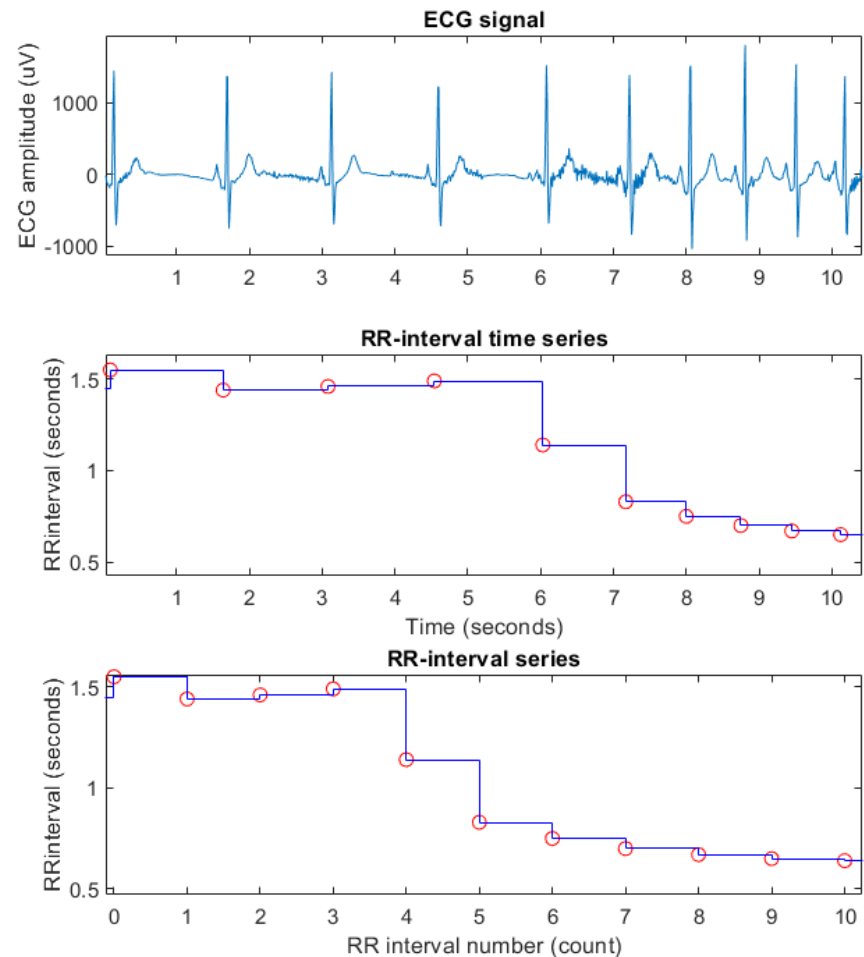
Method 1: resample RR time series

- We could resample time series evenly then apply DFT
- Advantages
 - conceptually easy
- Disadvantages
 - The resampled values depend on our interpolation method
 - The series has sudden changes (e.g. $t=6$ seconds). These cause unwanted high frequency components in PSD
- Method not recommended



Method 2: Spectrum of RR-intervals

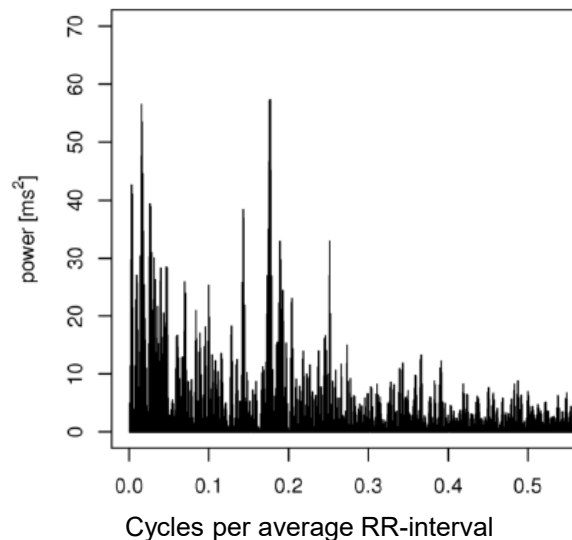
- We find the DFT of the RRinterval series (not the RR-interval time series)
- **Advantages**
 - No ambiguity about the method
- **Disadvantages**
 - the spectrum cannot be directly interpreted in terms of frequency.
 - Does need to deal with missing values



Calculations of HRV frequency domain features

– The interval based spectrum method via periodogram

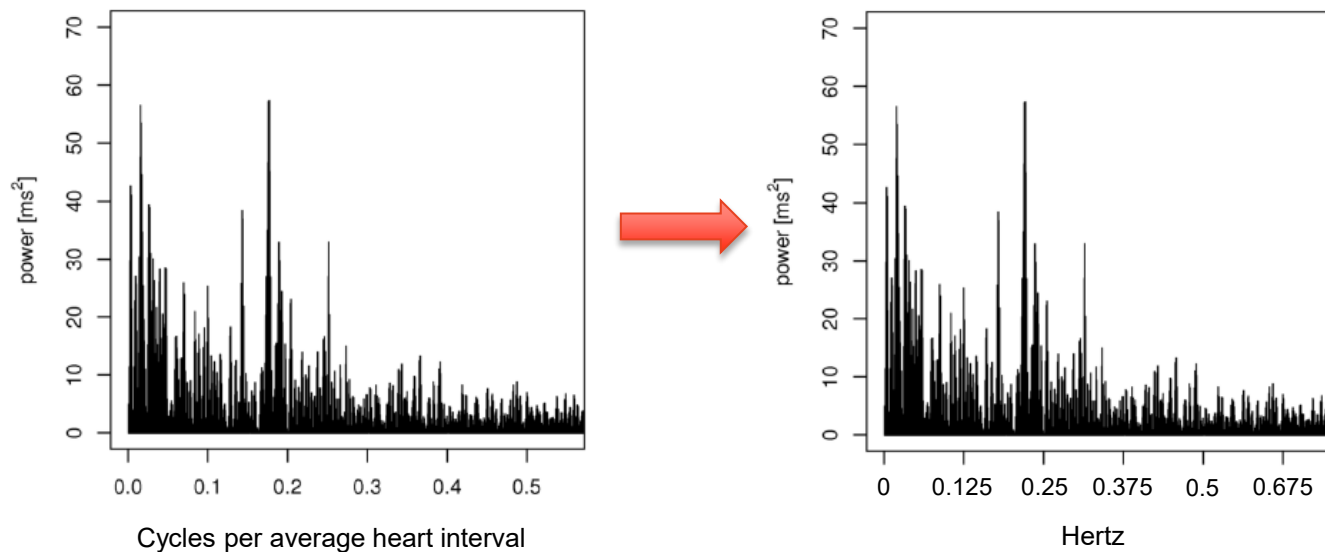
- Given the RR interval sequence RR
- we calculate the periodogram of the RR via the discrete fourier transform.
 - $|\mathcal{F}(RR - \text{mean}(RR))|^2$
 - The units of the x-axis is 'cycles per average RR-interval'
- Advantages: Easy to do e.g. in Matlab/[Python](#)
 - $P = \text{abs}(\text{fft}(RR))^2$
 - `np.abs(np.fft.fft(RR))**2`
- Disadvantages: If the RR interval values has missing values then cannot find the periodogram



*Only need lower
half of the
spectrum as upper
half is
symmetrical*

The interval based spectrum method

- **Converting from ‘cycles per average heart interval’ to Hertz**
 - Divide x-axis bin points by average RR-interval
- **Say average RR-interval is 0.8 seconds then**



The interval based spectrum method

Matlab Example: Major project, record 5

- *In the following example, we divide the ECG into into 1 minute segments, calculate a spectrum for each segment and then average the spectra*
- *After averaging , we scale spectrum to Hz.*

The interval based spectrum method, Matlab Example:

Major project, record 5

```
load ProjectTrainData.mat
i=5;
RR{5}=diff(QRSexpert{5})/100;
EpochLen=100*60;
close all
for j=1:length(ECG{5})/ EpochLen
    SegmentStart=(j-1)*EpochLen+1;
    SegmentEnd=j*EpochLen;
    RRseg=RR{i}(QRSexpert{i}>=SegmentStart & QRSexpert{i}<=SegmentEnd);
    if ~any(isnan(RRseg))
        PSD(j,:)=abs(fft(RRseg-mean(RRseg),256).^2);
    else
        PSD(j,1:256)=nan;
    end
end

%Average spectrum in cycles per interval
PSDavg=mean(PSD,1,'omitnan');
clf; plot([0:255]/256,PSDavg)
hold on; plot([0:255]/256,PSDavg,'.')
xlabel('cycles per average heart interval'); ylabel('Power (relative units)')

%Average spectrum (lower half) in Hertz
figure; plot([0:127]/256/mean(RR{i},'omitnan'),PSDavg(1:128))
hold on; plot([0:127]/256/mean(RR{i},'omitnan'),PSDavg(1:128),'.')
xlabel('Hertz');ylabel('Power (relative units)')
title(['Average heart interval = ',num2str(mean(RR{i}, 'omitnan')), ' seconds'])
```

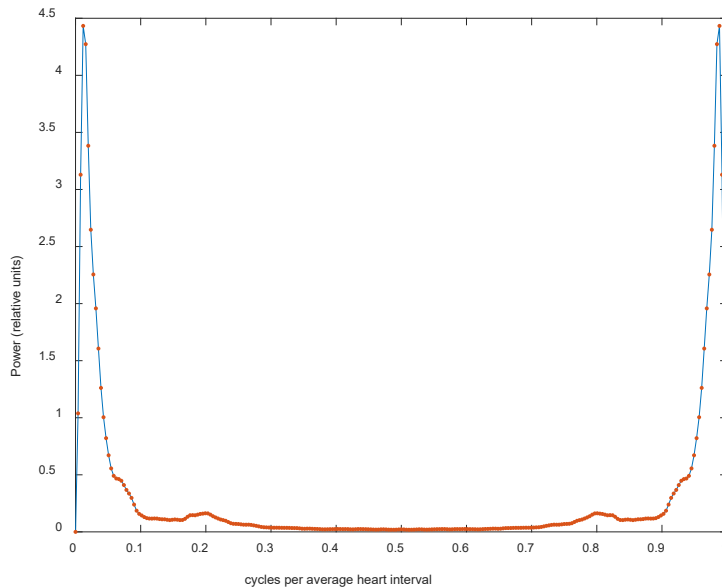
We divide ECG signal into 1 minute segments, calculate PSDs and average

If any NaNs in the segment can't calculate a spectrum for that segment

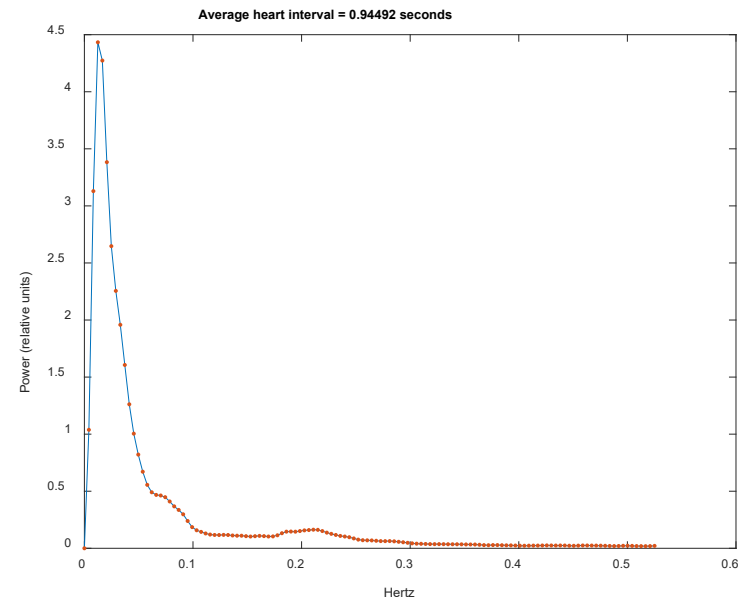
This zero pads the RR interval sequence to 256 points. All segment PSDs are same length

The interval based spectrum method

Matlab Example: record 5 for Major project data



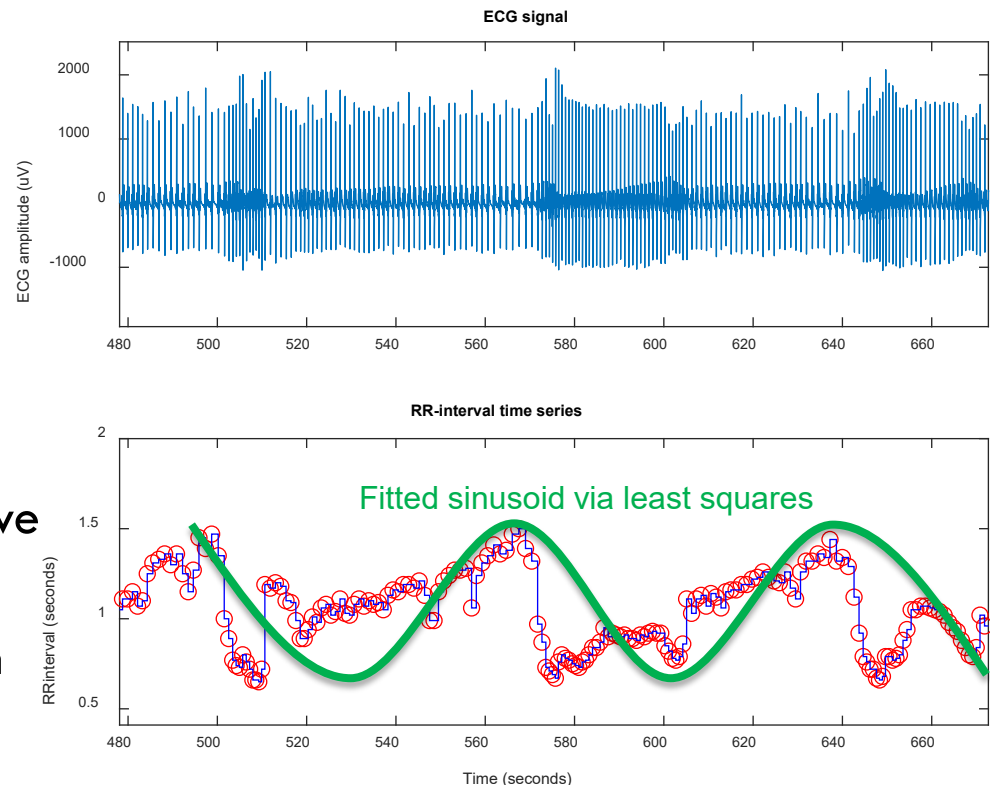
Full spectrum – note symmetry



Lower 1/2 spectrum scaled so x-axis is Hertz

Method 2: Lomb periodogram

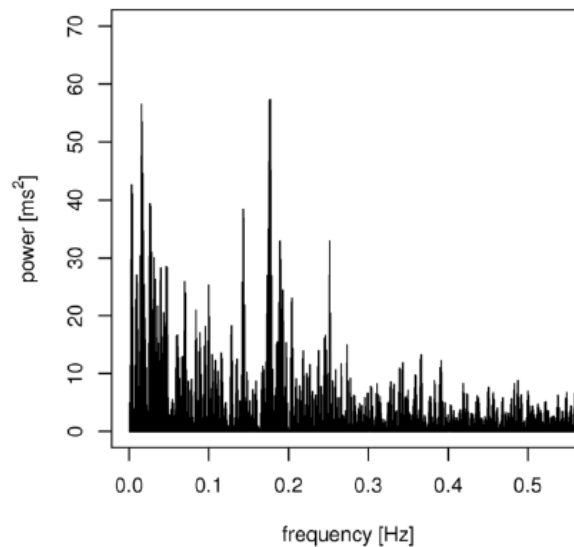
- The Lomb periodogram does not use the DFT but instead fits sinusoids to the data via least squares
- **Advantages**
 - Does not require evenly sampled data
 - Deals with missing data
- **Disadvantages**
 - More computationally intensive
 - $O(N^2)$
- **It is equivalent to Periodogram when**
 - Data is evenly sampled
 - The signal has zero mean
 - No missing data
- **Non-weighted least squares**



Calculations of HRV frequency domain features

– Lomb-Scargle periodogram

- Given the RR interval sequence RR we calculate the lomb periodogram of the RR . Lomb periodogram is based on least squares fitting of sinusoids to the data
- Advantages: can obtain a spectrum when there are missing values. Properly deals with the uneven sampling inherently involved in RR series
- Disadvantages: More complicated than periodogram above
- In Matlab we use *lomb* function.
- In Python we use *LombScargle* from *astropy.timeseries* `import LombScargle`



Lomb Scargle periodogram – Matlab code for Major Project data

```
load ProjectTrainData.mat
i=5;
RR{5}=diff(QRSexpert{5})/100;
EpochLen=100*60;
%Lomb-scargle PSD
clear PSD
figure
for j=1: length(ECG{i})/ EpochLen
    SegmentStart=(j-1)*EpochLen+1;
    SegmentEnd=j*EpochLen;
    RRseg=RR{i}(QRSexpert{i}>=SegmentStart & QRSexpert{i}<=SegmentEnd);
    QRSseg=QRSexpert{i}(QRSexpert{i}>(j-1)*EpochLen & QRSexpert{i}<=j*EpochLen)/100; %QRS times in seconds

    %Remove instances of identical QRS detection times
    idx=find(diff(QRSseg)==0);
    QRSseg(idx)=[];
    RRseg(idx)=[];

    if sum(RRseg)>30
        PSD(j,:)=plomb(RRseg,mean(RRseg,'omitnan'),QRSseg,'psd',0.5*[0:99]/100); %100 bins between 0 and 0.5 Hz
    else
        PSD(j,:)=nan(1,100);
    end
end

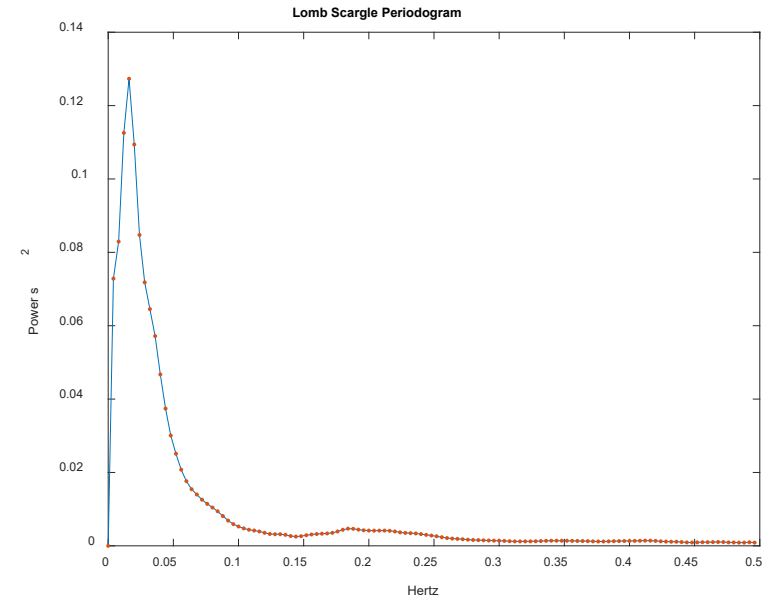
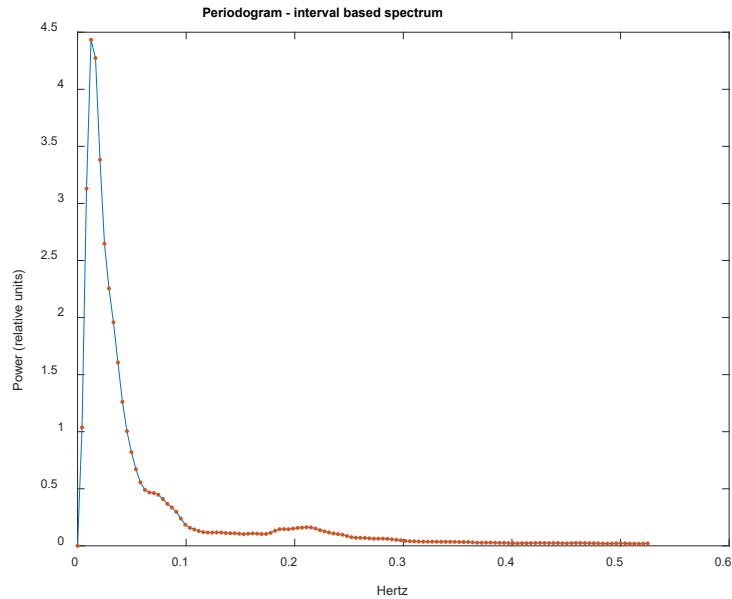
%Average spectrum in cycles per interval
PSDavg=mean(PSD,1,'omitnan');
clf; plot(0.5*[0:99]/100,PSDavg)
hold on; plot(0.5*[0:99]/100,PSDavg,'.')
xlabel('Hz'); ylabel('Power (relative units)')
```

We divide ECG signal into 1 minute segments, calculate PSDs and average

There seems to be instances of identical QRS detection times

Need a minimum coverage of 30 seconds by RR intervals before calculating spectrum

Comparison of RR spectra, Major project, record 5





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